FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness

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Abstract

Transformers are slow and memory-hungry on long sequences, as the time and memory complexity of self-attention are quadratic in sequence length. Approximate attention methods trade off model quality to reduce compute complexity, but often do not achieve wall-clock speedup. We propose FlashAttention, an IO-aware exact attention algorithm that uses tiling to reduce the number of memory reads/writes between GPU high bandwidth memory (HBM) and GPU on-chip SRAM.

1. Hardware Context & IO Bottleneck

Modern GPUs possess large High Bandwidth Memory (HBM, e.g. 40-80GB with ~1.5-2.0 TB/s bandwidth) and small, extremely fast SRAM (e.g. 192KB per Streaming Multiprocessor with ~19 TB/s bandwidth). Standard attention computes S = Q*K^T, P = softmax(S), and O = P*V, repeatedly reading and writing intermediate N x N matrices to slow HBM.

2. Algorithm: Tiling and Online Softmax
FlashAttention computes exact attention with O(N) memory footprint and substantially fewer HBM accesses by leveraging two core techniques:

1. Tiling: FlashAttention splits the input matrices Q, K, V into blocks, loads blocks into fast SRAM, computes attention with respect to that block, and incrementally updates the output.
2. Online Softmax: Softmax normally requires computing the denominator sum across the entire sequence before normalizing. FlashAttention uses online softmax scaling to incrementally update the running maximum and normalization sum without ever materializing the full N x N attention matrix in HBM.

Recomputation in Backward Pass: Rather than saving the intermediate attention matrix for backpropagation, FlashAttention stores only the softmax normalization statistics (LSE) and quickly recomputes attention on SRAM during the backward pass, dramatically slashing memory usage.